Article

Uncertainty-Aware Closed-Loop Model Predictive Control with EKF-SLAM for Safe Navigation of Nonholonomic Mobile Robots

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Abstract

Simultaneous Localization and Mapping (SLAM) supplies the state estimates that a model predictive controller (MPC) needs, yet the two are usually combined under a certainty-equivalence assumption that discards the uncertainty SLAM explicitly quantifies. This paper presents a closed-loop framework in which an Extended Kalman Filter SLAM (EKF-SLAM) estimator feeds a constrained nonlinear MPC for a unicycle robot navigating an *unknown* environment: landmarks and obstacles enter the estimator only once detected within a finite sensing range, and the posterior covariances of both the robot pose and the sensed obstacles size the keep-out constraints online through a chance-constraint-inspired, covariance-scaled safety margin. On a multi-leg patrol in Monte-Carlo simulation, SLAM reduces terminal error by 48% relative to odometry while building the map from scratch to centimetre accuracy. Against a sensed static obstacle, fixed margins collide in 36% (odometry) and 14% (SLAM) of trials, whereas the covariance-aware margin eliminates collisions (0 of 50) for less than 1% additional path. An ablation shows a fixed margin of the same average size is equally safe, locating the covariance-aware shaping's value in online self-tuning. The mechanism extends to an intermittently visible moving obstacle, where it again matches the clairvoyant oracle provided stale tracks are dropped. A scaling-factor sweep characterizes the safety–efficiency trade-off.

Keywords: mobile robotics; EKF-SLAM; model predictive control; obstacle avoidance; dynamic obstacle avoidance; covariance-scaled safety margins; uncertainty-aware control; nonholonomic systems

1. Introduction

Over the past two decades, interest in autonomous mobile robots has surged, driven by their potential across logistics, assistive care, and service operations. To act effectively in such settings, a robot must simultaneously ascertain its own location, build or exploit a map of its surroundings, follow a desired motion, and evade obstacles—all the while contending with sensor noise and actuator limits. Two technologies dominate the perception and control halves of this problem. Simultaneous Localization and Mapping (SLAM), and in particular Extended Kalman Filter SLAM (EKF-SLAM), furnishes a recursive, probabilistic estimate of the robot pose and landmark map together with its covariance [1,2]. Model Predictive Control (MPC), in turn, methodically anticipates the consequences of prospective actions, optimizing the control inputs over a receding horizon while respecting constraints [3].

Received:
Revised:
Accepted:
Published:

 https://doi.org/10.3390/s1010000

In practice, however, these two components are designed in isolation and stitched together under a *certainty-equivalence* assumption: the controller embraces the SLAM estimate as though it were the true state, and the localization *uncertainty* —which the filter already computes explicitly—is quietly discarded. Such an assumption is benign while the estimates remain accurate, yet it becomes precisely the wrong one in the vicinity of obstacles, where a confident controller acting on an uncertain estimate may steer the *true* robot into a collision while believing itself perfectly safe.

This paper sets out to close that gap in the setting SLAM was conceived for: an environment that is *unknown before it is sensed*. The map is not given—each landmark enters the estimator only once the robot first detects it within a finite sensing range—and the obstacles are likewise discovered, not declared: their positions (and, for a moving obstacle, velocities) are estimated online from the robot's own detections. We integrate this exploratory EKF-SLAM with a constrained nonlinear MPC for a unicycle robot and, rather than halting at certainty-equivalence, we feed the *covariances*—of the pose and of every sensed obstacle—back into the controller, so that the keep-out constraints widen when the robot is unsure of itself or of what it has seen, and tighten as its confidence returns. The contributions of this study are as follows:

1. An end-to-end EKF-SLAM $\rightarrow$ NMPC pipeline for the unicycle model in an unknown environment: landmarks initialized from first detection under a finite sensing range, obstacles sensed and estimated online, and a genuinely constrained controller (actuator limits and obstacle avoidance) solved by a multiple-shooting transcription.
2. A *covariance-aware* obstacle-avoidance formulation: a chance-constraint-inspired safety margin that inflates each keep-out radius by a term proportional to the square root of the dominant eigenvalue of the combined pose and obstacle-estimate covariance.
3. A controlled Monte-Carlo study on a multi-leg patrol against an ideal-state oracle and a dead-reckoning baseline, quantifying tracking accuracy, map accuracy, collision rate, and the safety–efficiency trade-off.
4. An extension to *moving*, intermittently visible obstacles via a constant-velocity EKF tracker that coasts when the obstacle leaves the sensing range, including the track management this necessitates: without dropping stale tracks, the unboundedly growing prediction covariance trades mission completion for safety.
5. An ablation that isolates margin *size* from *adaptivity*, showing the collision-free level is set by the average margin while the covariance-aware shaping's distinct value is that it self-tunes the keep-out online (no hand-tuned constant) and is modestly more path-efficient.

2. Related Work

MPC for nonholonomic robots. Receding-horizon control of wheeled mobile robots is by now a well-trodden field. Howard et al. [4] demonstrated model-predictive control that threads a robot along intricate paths under demanding mobility constraints, while Wang [5] lays out the systematic design of linear time-varying MPC about a reference trajectory. For the unicycle model in particular, Mehrez et al. [6] showed that a stabilizing nonlinear MPC (NMPC) can be solved online with open-source real-time optimization; it is this NMPC formulation that we adopt and extend with obstacle-avoidance constraints.

Output-feedback and estimation-in-the-loop. When the state cannot be measured directly, MPC is paired with an observer or filter and joined to it under certainty equivalence [3]. Robust and tube MPC instead confront the estimation and disturbance error head-on, tightening the constraints over a disturbance-invariant tube so as to guarantee their satisfaction, though at the price of constructing invariant sets [7]. Planning-under-uncertainty methods such as LQG-MP [8] press further still, characterizing in advance the a priori state distribution that a chosen controller and sensor model induce, and selecting paths of low collision probability before execution ever begins. Our own approach is lighter than tube MPC and, unlike the offline analysis of LQG-MP, adapts *online*: it reuses the covariance that the estimator already maintains and brings it to bear only on the obstacle constraints that dominate safety.

SLAM and active/perception-aware planning. EKF-SLAM is a classical instrument of probabilistic robotics that maintains the joint pose–map covariance [1,2], and the broader arc of SLAM research is surveyed by Cadena et al. [9]. Active SLAM and perception-aware control—the perception-aware MPC of Falanga et al. [11] among them—reshape the robot's motion or its control objective in the service of better perception and estimation. Our stance is different: we neither alter the estimation objective nor chase information gain, but simply take the covariance the filter already yields and press it into service to make the *controller* safer, leaving the estimator untouched and confining the contribution to the control–constraint interface.

Chance-constrained MPC. Probabilistic (chance) constraints translate a bound on collision probability into a deterministic tightening of the constraint, by a multiple of the state-uncertainty standard deviation; Mesbah [12] surveys the stochastic-MPC landscape, and Blackmore et al. [13] approximate chance-constrained predictive control through particle and sampling methods. Nearest to our own obstacle treatment, Zhu and Alonso-Mora [14] formulate a real-time chance-constrained NMPC for collision avoidance in which the uncertainty springs from the predicted motion of dynamic obstacles. We embrace the same constraint-tightening device but drive it from a different and underused source—the robot's own online EKF-SLAM covariance, combined with the covariance of each obstacle estimate it maintains—so that the safety margin follows the live quality of both localization and perception rather than worst-case bounds.

3. System Models

3.1. Unicycle Kinematic Model

The robot is modeled as a unicycle with pose $\mathbf{x}_{r,t} = [x_t, y_t, \theta_t]^\top$ and control $\mathbf{u}_t = [v_t, \omega_t]^\top$, where v_t is the linear velocity and ω_t the angular velocity. The continuous-time kinematics are

$$\dot{x} = v\cos\theta, \qquad \dot{y} = v\sin\theta, \qquad \dot{\theta} = \omega. \tag{1}$$

For control we use the Euler discretization at sampling period Δt,

$$\mathbf{x}_{r,t+1} = \mathbf{x}_{r,t} + \Delta t \begin{bmatrix} v_t\cos\theta_t \\ v_t\sin\theta_t \\ \omega_t \end{bmatrix}, \tag{2}$$

while for the EKF-SLAM prediction we use the exact integral of the constant-twist motion over Δt, which avoids the linear-arc error of (2) for $\omega \neq 0$.

3.2. Range–Bearing Measurement Model

The environment contains M point landmarks with coordinates $(l_{x,i}, l_{y,i})$. The robot observes each landmark through a range–bearing sensor,

$$\mathbf{z}_t^i = \begin{bmatrix} d_t^i \\ \phi_t^i \end{bmatrix} = \begin{bmatrix} \sqrt{(l_{x,i}-x_t)^2 + (l_{y,i}-y_t)^2} \\ \operatorname{atan2}(l_{y,i}-y_t,\ l_{x,i}-x_t) - \theta_t \end{bmatrix} + \mathbf{v}_t^i, \qquad \mathbf{v}_t^i \sim \mathcal{N}(\mathbf{0}, \mathbf{R}), \tag{3}$$

with $\mathbf{R} = \operatorname{diag}(\sigma_d^2, \sigma_\phi^2)$. Bearings are wrapped to $(-\pi, \pi]$. The sensor has a finite range r_{sense}: a landmark (or obstacle) is observed only while it lies within r_{sense} of the robot, so the environment is genuinely *unknown before it is detected*. Landmarks are assumed distinguishable (known data association), as with uniquely identifiable features or beacons.

4. Methodology

The augmented EKF-SLAM state stacks the robot pose and the M landmarks, $\mathbf{x}_t = [\mathbf{x}_{r,t}^\top, l_{x,1}, l_{y,1}, \ldots, l_{x,M}, l_{y,M}]^\top \in \mathbb{R}^{3+2M}$, estimated by mean $\hat{\mathbf{x}}_t$ and covariance $\mathbf{P}_t$.

4.1. EKF-SLAM

We use the standard EKF-SLAM formulation [1]. At each step the filter performs a prediction using the commanded control $\mathbf{u}_t$ (the filter does not observe the actuation noise, so process uncertainty is genuine) followed by a measurement update. The prediction propagates the pose with the exact velocity-motion model and inflates the covariance through the control-to-state Jacobian V_t and input-noise covariance $\mathbf{Q}_u = \mathrm{diag}(\sigma_v^2, \sigma_\omega^2)$:

$$\hat{\mathbf{x}}_t^- = \hat{\mathbf{x}}_{t-1} + F_x^\top\, g(\hat{\mathbf{x}}_{t-1}, \mathbf{u}_t), \qquad \mathbf{P}_t^- = G_t\, \mathbf{P}_{t-1}\, G_t^\top + F_x^\top\, (V_t \mathbf{Q}_u V_t^\top)\, F_x, \tag{4}$$

where F_x maps the pose block into the full state and G_t is the motion Jacobian. The update is initialized with the prior, $\hat{\mathbf{x}}_t \leftarrow \hat{\mathbf{x}}_t^-$ and $\mathbf{P}_t \leftarrow \mathbf{P}_t^-$, and landmarks are processed *sequentially*: for each observed landmark i the predicted measurement $\hat{\mathbf{z}}_t^i$ and Jacobian H_t^i are computed from the current *estimated* landmark (no ground-truth information is used), the innovation bearing is wrapped, and the standard Kalman correction is applied before processing the next landmark:

$$K_t^i = \mathbf{P}_t H_t^{i\top} \left(H_t^i\, \mathbf{P}_t\, H_t^{i\top} + \mathbf{R}\right)^{-1}, \quad \hat{\mathbf{x}}_t \leftarrow \hat{\mathbf{x}}_t + K_t^i\big(\mathbf{z}_t^i - \hat{\mathbf{z}}_t^i\big), \quad \mathbf{P}_t \leftarrow \big(I - K_t^i H_t^i\big)\, \mathbf{P}_t. \tag{5}$$

The map is *unknown a priori*. Each landmark is initialized from its first in-range observation through the inverse measurement model, $\hat{l}_{x,i} = \hat{x} + d\cos(\hat{\theta} + \phi)$, $\hat{l}_{y,i} = \hat{y} + d\sin(\hat{\theta} + \phi)$, under an uninformative prior, and is refined by every subsequent re-observation; landmarks outside the sensing range contribute no update, so the pose covariance grows through landmark-poor stretches and contracts on re-detection. We adopt the standard *start-frame* convention of exploratory SLAM: the robot's initial pose defines the world frame and carries zero initial uncertainty. This fixes the gauge freedom of unknown-map SLAM—a rigid transformation of robot and map together leaves every range–bearing measurement unchanged, so a pose in an externally defined frame is unobservable—and it makes navigation goals well posed as coordinates *relative to the deployment point*, which is how a robot without external infrastructure operates in practice. In this setting the benefit of SLAM over dead reckoning is *drift suppression*: odometric error grows without bound along the mission, whereas re-observing self-mapped landmarks bounds it.

4.2. Constrained Nonlinear MPC

At each sampling instant the controller solves, over horizon N,

$$\begin{aligned}
\min_{\mathbf{X},\mathbf{U}} \; & \sum_{k=0}^{N-1} \Big[(\mathbf{x}_{r,k} - \mathbf{x}^\star)^\top \mathbf{W}_x (\mathbf{x}_{r,k} - \mathbf{x}^\star) + \mathbf{u}_k^\top \mathbf{W}_u\, \mathbf{u}_k \Big] \\
\text{s.t. } & \mathbf{x}_{r,0} = \hat{\mathbf{x}}_{r,t}, \qquad \mathbf{x}_{r,k+1} = \mathbf{x}_{r,k} + \Delta t\, f(\mathbf{x}_{r,k}, \mathbf{u}_k), \\
& v_{\min} \le v_k \le v_{\max}, \quad \omega_{\min} \le \omega_k \le \omega_{\max}, \\
& \|\mathbf{p}_k - \mathbf{o}_j\| \ge r_{\text{rob}} + r_j + \delta, \quad j = 1, \ldots, N_o,\; k = 0, \ldots, N,
\end{aligned} \tag{6}$$

where $\mathbf{x}^\star$ is the active reference posture, $\mathbf{x}_{r,k}$ the predicted pose over the horizon (index k), $\mathbf{p}_k = [x_k, y_k]^\top$, $\mathbf{o}_j$ and r_j the j-th obstacle center and radius, r_{rob} the robot radius, and

$\delta \geq 0$ a safety margin defined below. The initial state is set to the pose block $\hat{\mathbf{x}}_{r,t}$ of the current EKF-SLAM estimate. Missions are specified as an ordered list of waypoints in the start frame; a simple mission executive supplies the active waypoint as $\mathbf{x}^\star$, advancing when the estimate arrives within a pass-through tolerance or a per-waypoint timeout expires (the timeout prevents an erroneous estimate hovering just outside the tolerance from wedging the mission; the estimation error still registers in the terminal metric). The problem is transcribed by *multiple shooting* (states and controls are decision variables, dynamics enter as equality constraints), which improves convergence of the nonconvex obstacle constraints, and solved with an interior-point method [6]. Only the first control is applied; the horizon is then shifted and warm-started.

4.3. Closed-Loop Coupling

The closed loop runs at period Δt: EKF-SLAM produces $(\hat{\mathbf{x}}_t, \mathbf{P}_t)$; the NMPC (6) consumes the pose estimate $\hat{\mathbf{x}}_{r,t}$ (and, below, $\mathbf{P}_t$) and returns $\mathbf{u}_t$; the actuated control $\mathbf{u}_t + \mathbf{w}_t$ drives the true plant; the new range–bearing measurement updates the filter. Under certainty equivalence the controller would use $\hat{\mathbf{x}}_{r,t}$ only; our contribution additionally uses $\mathbf{P}_t$.

4.4. Sensed Obstacles and the Covariance-Aware Safety Margin

Consistent with the unknown-map setting, obstacles are not declared to the controller—they are *discovered*. A static obstacle is estimated by a two-state EKF driven by the robot's range–bearing detections: the estimate $\hat{\mathbf{o}}$ is initialized from the first in-range detection through the inverse measurement model, with an initial covariance Σ_o that inherits the robot's pose uncertainty through the corresponding Jacobians, and each subsequent update folds the current pose covariance into an effective measurement noise $\mathbf{R}_{\text{eff}} = \mathbf{R} + H_p \mathbf{P}_{\text{pose}} H_p^\top$, keeping Σ_o honest without a joint filter. Until the obstacle is first detected, the keep-out constraint in (6) is inactive: the controller cannot avoid what it has not yet sensed, and the sensing range must therefore afford sufficient reaction distance.

A fixed margin $\delta = \delta_0$ cannot account for time-varying estimation quality. We therefore size the margin from the live covariances. Let $\mathbf{P}_{xy}$ be the 2×2 position block of $\mathbf{P}_t$, Σ_o^{pos} the position block of the obstacle-estimate covariance, and $\lambda_{\max}(\cdot)$ the largest eigenvalue. We set

$$\delta = \delta_0 + \gamma \sqrt{\lambda_{\max}\big(\mathbf{P}_{xy} + \Sigma_o^{\text{pos}}\big)}, \tag{7}$$

where δ_0 is a fixed buffer (chosen so that an ideal-state controller is collision-free despite inter-sample discretization) and $\gamma \geq 0$ scales the tightening in the manner of chance-constrained MPC, in which constraints are tightened by a multiple of the state standard deviation to bound the violation probability [12,13]. We stress that (7) is chance-constraint-*inspired* rather than a calibrated probability bound: linearization error, non-Gaussianity, and correlations across horizon nodes preclude an exact guarantee, so γ is selected empirically from the safety–efficiency sweep reported in the Results. The square-root term is the relative-position uncertainty along its worst-case direction, so the keep-out radius grows precisely when, and in the direction in which, the robot is least sure of itself or of the obstacle. Equation (7) is available only because the filters provide their covariances online; a dead-reckoning controller has no comparable, trustworthy uncertainty to act on.

4.5. Extension to Dynamic Obstacles and Track Management

The framework extends to a *moving* obstacle without changing the estimator or controller structure. The obstacle is tracked by a constant-velocity (CV) EKF from the robot's range–bearing detections, producing a position–velocity estimate $\hat{\mathbf{o}}_t$ and covariance $\Sigma_{o,t}$, in the spirit of chance-constrained dynamic avoidance [14]. As with the static case, the track is created on the first in-range detection with a covariance that inherits the pose uncertainty (the initial velocity is unknown and receives a large prior). Because the sensing range is finite, visibility is *intermittent*: while the obstacle is out of range the tracker *coasts*, predicting without correction, and its covariance grows. Over the horizon the obstacle is propagated by the CV model to $\hat{\mathbf{o}}_{t+k|t}$ with covariance $\Sigma_{o,k}$ that *grows* with the look-ahead k. The keep-out constraint in (6) becomes time-varying,

$$\|\mathbf{p}_k - \hat{\mathbf{o}}_{t+k|t}\| \geq r_{\text{rob}} + r_{\text{obs}} + \delta_k, \tag{8}$$

and the covariance-aware margin combines both uncertainty sources,

$$\delta_k = \delta_0 + \gamma\sqrt{\lambda_{\max}\big(\mathbf{P}_{xy} + \Sigma^{\text{pos}}_{o,k}\big)}, \tag{9}$$

where $\Sigma^{\text{pos}}_{o,k}$ is the position block of the predicted obstacle covariance. The keep-out bubble thus inflates both because the robot is unsure of its own pose and because the obstacle's future is uncertain, more so further ahead. To preserve feasibility in tight encounters, the obstacle constraint is softened with a slack variable penalized in the cost.

Intermittent visibility makes one further component *necessary*: track management. A track that has not been re-detected for a staleness window is dropped, and the constraint deactivates until re-detection. Without it, the coasting covariance grows without bound and (9) inflates a stale phantom bubble that can blockade the workspace: in our experiments the un-managed tracker kept collisions near zero but left the mission unfinished in many trials. This is not an incidental implementation detail but a structural property of uncertainty-aware margins under partial observation—the margin is only as reasonable as the lifetime of the estimate that feeds it.

In summing the two covariances in (9) we treat the robot and obstacle uncertainties as independent. The two are in fact coupled—the obstacle is detected from the robot's own uncertain pose—but this coupling correlates the two errors *positively*, and positively correlated errors partially cancel in the relative position; the sum of the marginal covariances therefore over-approximates the relative-position uncertainty, so the approximation errs on the conservative side. A joint robot–obstacle filter would recover the cross-covariance at the price of added complexity.

5. Simulation Setup

Scenario. The robot executes a return-to-start *patrol*: from the start pose $(0,0,0)$, which defines the world frame, it visits the waypoints $(1.5,1.5)$ and $(-1.5,1.0)$ and returns to the origin, a mission of roughly 7 m. Three landmarks—unknown to the robot beforehand—lie at $(-0.5,0.5)$, $(1,1)$, and $(-1.5,0)$; the sensing range is $r_{\text{sense}} = 1.2$ m, chosen so that sensing coverage genuinely varies along the mission: only the first landmark is in range at the start and the rest are discovered en route, while the middle of the second leg is landmark-poor and the pose covariance swells there—the nonstationarity the covariance-aware margin exists to respond to. For the static-obstacle case a disk of radius 0.15 m sits at $(0,1.25)$, on the second leg—encountered after drift has accumulated, and itself unknown until detected. The robot radius is 0.15 m. Actuator limits are $v_{\max} = 0.6$ m/s, $\omega_{\max} = \pi/4$ rad/s. The free-space controller uses $N = 10$, $\Delta t = 0.05$ s; the obstacle controller uses $N = 14$, $\Delta t = 0.2$ s for adequate lookahead. Cost weights are $\mathbf{W}_x = \text{diag}(1,5,0.1)$, $\mathbf{W}_u = \text{diag}(0.5,0.05)$. The mission executive uses a 0.10 m pass-through tolerance and a per-waypoint timeout for intermediate waypoints and a 0.05 m tolerance at the final goal. The timeout is 15 s in the static studies and 30 s (with an extended 75 s mission budget) in the dynamic scenario: a safe controller must be allowed to wait out a crossing obstacle rather than being forced to abandon a temporarily blocked waypoint, while the timeout still bounds the pathological case of an erroneous estimate hovering just outside the arrival tolerance.

Noise. Range and bearing measurement noise have variance $\sigma_d^2 = \sigma_\phi^2 = 0.01$ (standard deviations 0.1 m and 0.1 rad); actuation noise on v and ω has variance 0.01 (standard deviation 0.1), a level at which dead-reckoning drift is material over the mission—the regime SLAM exists for. The filter is given the same noise parameters (a consistent filter). There is no initial pose offset: the start pose defines the frame and carries zero uncertainty, so all localization error is *accumulated*, not injected. The fixed buffer is $\delta_0 = 0.06$ m and the scaling factor is $\gamma = 2$ unless swept. In the dynamic scenario the obstacle starts at $(1.3, 0.15)$ with crossing velocity $(-0.16, 0.16)$ m/s and true acceleration noise of standard deviation 0.05, a moderately manoeuvring adversary whose speed remains below the robot's over the 75 s mission; the tracker assumes a larger acceleration variance (0.02), so its coasting predictions err on the conservative side. Tracks are dropped after 3 s without a detection: by then the coasting position uncertainty (≈0.4 m at the assumed acceleration noise) exceeds the physical keep-out radius, and the prediction is no longer actionable.

Baselines. Feedback sources and obstacle strategies are compared on *identical* noise realizations per trial: *oracle* (the NMPC is given the true pose and, in obstacle cases, the true obstacle; an upper performance bound), *odom* (dead reckoning: EKF prediction only, no measurement correction), and *slam* (full EKF-SLAM). Sharing the noise across modes isolates the effect of the estimator or margin under test.

Metrics. We report the *terminal true-vs-reference error* (the distance between the *true* final pose and the final waypoint, that is, what the robot actually achieves, not what it believes), the mean localization error, the final *map error* (mean distance between estimated and true landmark positions), the collision rate and minimum obstacle clearance judged on true geometry, and the path length. We run $N_{\mathrm{MC}} = 50$ Monte-Carlo trials and report means with 95% confidence intervals (Wilson intervals for collision proportions). All collision statistics pertain to the scenario geometries described above, on which γ is also selected by the sweeps; transfer of the collision-free operating points to unseen layouts is not claimed.

Implementation. The framework is implemented in MATLAB. The constrained NMPC is formulated and solved with CasADi [15] (v3.5.5) using its bundled IPOPT interior-point solver, whereas the EKF-SLAM estimator, the static obstacle estimator, and the constant-velocity tracker are implemented directly in MATLAB.

6. Results

6.1. Tracking and Mapping on the Free-Space Patrol

We begin without obstacles. Table 1 and Figure 1 gather the patrol study. Dead reckoning drifts as the mission lengthens, ending 0.122 m from the final waypoint with a mean localization error of 0.083 m; closing the loop with EKF-SLAM pulls the terminal error down to 0.064 m—a 48% reduction, with non-overlapping confidence intervals—and halves the localization error to 0.041 m, while simultaneously building the landmark map *from scratch* to a final accuracy of 0.040 m. SLAM lands within 0.017 m of the oracle bound. The uncertainty trace (Figure 1, right) shows the signature this framing is designed to exhibit: the pose covariance swells through the landmark-poor mid-tour and contracts on each re-detection, whereas the dead-reckoning covariance only grows. The return-to-start leg additionally re-observes the landmarks mapped at the outset—a loop closure in miniature—which is what pulls SLAM's terminal error back to the oracle bound while odometry, having no such mechanism, retains its accumulated drift.

Table 1. Free-space patrol Monte-Carlo results ($N_{\mathrm{MC}} = 50$, mean $\pm$ 95% CI). Map error is the final mean landmark-position error (SLAM builds the map online; the others maintain none).

Feedback	Terminal Error [m]	Localization Error [m]	Map Error [m]
oracle	0.0466 ± 0.0008	0.0000	—
odom	0.1224 ± 0.0179	0.0829 ± 0.0105	—
slam	$\mathbf{0.0641 \pm 0.0093}$	$\mathbf{0.0410 \pm 0.0059}$	**0.0403**

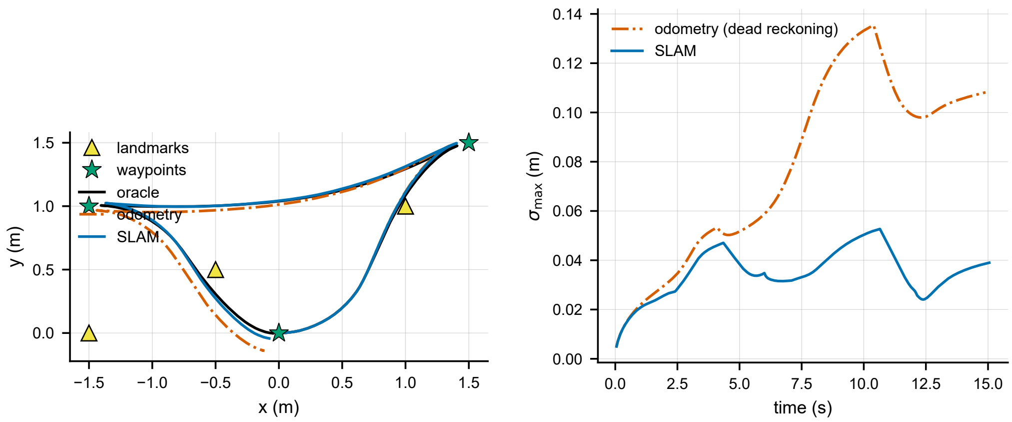

Figure 1. Free-space patrol. (**Left**) true trajectories for one trial; (**Right**) pose-uncertainty evolution $\sigma_{\max}$ over the tour: SLAM's uncertainty contracts on each landmark re-detection while dead reckoning's only grows.

6.2. Sensed Static Obstacle: Fixed vs. Covariance-Aware Margin

We now place the obstacle on the landmark-poor second leg, unknown to the robot until detected. With a *fixed* margin δ_0, the clairvoyant oracle never collides, but odometry collides in 36% of trials (worst penetration -0.171 m) and SLAM in 14% (Table 2). SLAM's better estimate helps—its collision rate is less than half of odometry's, and its obstacle estimate converges to 0.068 m—but a fixed margin cannot exploit the fact that SLAM *knows* when it is uncertain, and the passage happens precisely where the pose covariance has swollen.

Activating the covariance-aware margin (7) with $\gamma = 2$ eliminates SLAM collisions entirely (0% over 50 trials, worst clearance $+0.073$ m) while leaving the terminal accuracy unchanged (0.089 m vs. 0.091 m). Notably, the safety is almost free: the mean path grows by less than 1% (6.99 m vs. 6.93 m), because the margin concentrates its inflation where and when uncertainty is high—the landmark-poor passage—rather than paying a detour everywhere. Because the noise seeds are shared, the oracle and odometry columns are identical across margin settings, confirming that the only change is the margin. Figure 2 shows the trajectories and the margin dynamics: the applied inflation visibly swells as the robot enters the landmark-poor stretch and relaxes as coverage returns.

Table 2. Sensed-obstacle Monte-Carlo results ($N_{MC} = 50$; Wilson 95% CIs). All non-oracle strategies estimate the obstacle online from detections.

Strategy	Collision Rate (95% CI)	Worst Clearance [m]	Terminal Error [m]	Path [m]
oracle (clairvoyant)	0% [0.0, 7.1]	+0.023	0.034	6.92
odom + fixed δ_0	36% [24.1, 49.9]	−0.171	0.201	6.85
slam + fixed δ_0	14% [7.0, 26.2]	−0.058	0.091	6.93
slam + covariance-aware	**0% [0.0, 7.1]**	**+0.073**	**0.089**	**6.99**

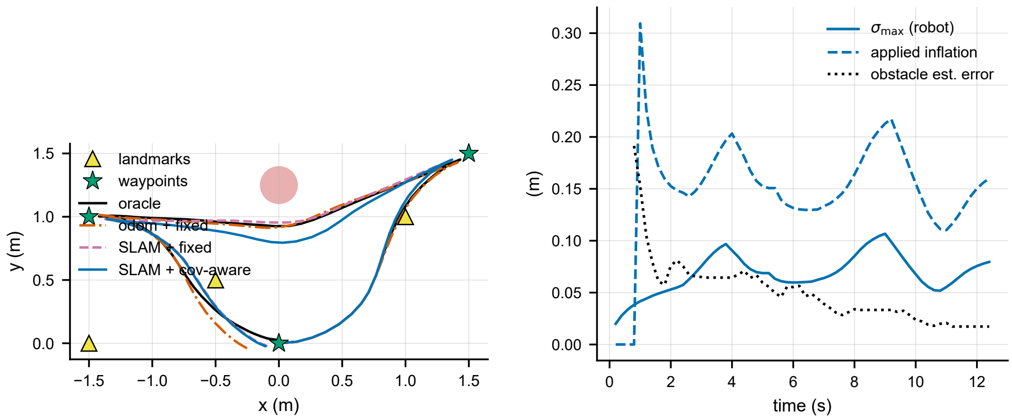

Figure 2. Sensed static obstacle. (**Left**) true trajectories for one trial; (**Right**) covariance-aware margin dynamics over the same trial: the applied inflation initializes at first detection and follows the pose and obstacle-estimate uncertainty, swelling through the landmark-poor passage.

6.3. Safety–Efficiency Trade-off

Figure 3 and Table 3 sweep the scaling factor γ. The collision rate falls monotonically from 14% at $\gamma = 0$ to 2% at $\gamma = 0.5$ and is 0% for every $\gamma \geq 1$, while the terminal accuracy is unaffected throughout (≈ 0.09 m). The striking feature of this sweep is how little the safety costs: the mean path grows by only 1.5% across the entire range (6.93 m at $\gamma = 0$ to 7.04 m at $\gamma = 3$), and the worst-case clearance climbs steadily (−0.058 to +0.141 m). Because the margin keys off the live covariance, it spends clearance only in the landmark-poor passage where the uncertainty actually resides; a fixed margin of comparable safety must pay everywhere. The sweep justifies the operating point $\gamma = 2$: it lies well inside the collision-free regime with headroom against sampling variability.

Table 3. Scaling-factor (γ) sweep ($N_{MC} = 50$, sensed static obstacle). $\gamma = 0$ is the fixed margin δ_0.

γ	Collision Rate	Path Length [m]	Terminal Error [m]	Worst Clearance [m]
0.0	14%	6.93	0.091	−0.058
0.5	2%	6.95	0.092	−0.026
1.0	0%	6.96	0.088	+0.007
1.5	0%	6.98	0.090	+0.042
2.0	0%	6.99	0.089	+0.073
2.5	0%	7.01	0.085	+0.106
3.0	0%	7.04	0.084	+0.141

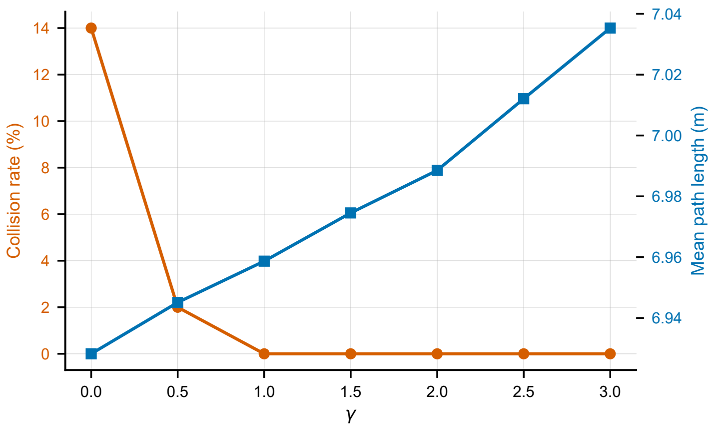

Figure 3. Safety–efficiency trade-off of the covariance-aware margin (sensed static obstacle). Collision rate (left axis) falls monotonically to zero as γ grows; mean path length (right axis) increases by less than 2% across the whole range.

6.4. Dynamic Obstacles under Intermittent Visibility

We now let the obstacle move, crossing the first patrol leg with random accelerations, so a constant-velocity prediction is necessarily imperfect—and, because the sensing range is finite, the obstacle is repeatedly *lost and re-acquired* as robot and obstacle wander apart and back. Four strategies are pitted against one another: *oracle* (true pose and clairvoyant true obstacle trajectory), *static* (freeze the obstacle at its current estimate), *cv_fixed* (CV-predicted track, fixed margin), and *cv_cov* (CV-predicted track, covariance-aware margin (9), $\gamma = 2$). Robot localization is EKF-SLAM in all non-oracle modes, so the comparison isolates the obstacle-handling strategy.

Intermittent visibility makes this scenario far harsher than a fully observed one: Table 4 shows that freezing the obstacle collides in 88% of trials and CV prediction with a fixed margin still collides in 44%—the prediction helps, but a fixed margin cannot absorb the error a coasting track accumulates. The covariance-aware margin eliminates collisions (0% over 50 trials, worst clearance $+0.210$ m), matching the clairvoyant oracle, at the cost of a substantially more cautious path (15.4 m vs. the oracle's 7.0 m): with only intermittent contact, caution is paid in detours.

The detours carry a second, subtler cost that the collision and terminal metrics do not expose, and Table 4 therefore reports *true* waypoint achievement: the number of waypoints the true robot passes within 0.15 m of (one robot radius), and the mean true miss distance. The covariance-aware controller strictly visits only 2.0 of 3 waypoints (mean miss 0.11 m), fewer than the colliding fixed strategies (2.4–2.5). The mechanism is not the executive abandoning blocked waypoints: in a representative missed trial the *estimate* arrived 0.07 m from the waypoint —within tolerance—while the true robot passed 0.48 m away, the localization error having grown to 0.41 m at that moment (against 0.04 m in the free-space patrol). The safety detour itself degrades localization: evading the obstacle takes the robot away from the landmarks through aggressive manoeuvres where drift compounds, and the waypoints are then visited in the drifted belief frame, with the estimate re-anchoring only on the homeward leg (hence the healthy terminal error). Crucially, the safety claim is unaffected—collisions are judged on true geometry, and the margin scales with precisely the covariance that grows during the detour, so it keeps covering the very error it causes. The fixed strategies post better waypoint numbers only because they do not detour—and they collide in 44–88% of trials doing so.

Read at the *mission* level, the comparison is therefore not close: a collided patrol is a failed patrol. The fixed strategies buy their extra waypoint precision on missions that end in collision 44–88% of the time, whereas the covariance-aware controller completes every patrol without contact and still passes within 0.11 m of its waypoints on average. Scored over collision-free missions only—the score an operator cares about—the covariance-aware controller dominates the comparison.

Track management proved *necessary*, not optional. Without the 3 s staleness drop, the covariance-aware controller remained nearly collision-free but its coasting track's covariance grew without bound, inflating a stale phantom bubble that blockaded the workspace; in an early configuration of this scenario (a more strongly manoeuvring obstacle, acceleration noise standard deviation 0.15, before the staleness drop was introduced) the mean terminal error ballooned beyond 2.5 m because many patrols simply could not be completed. Dropping stale tracks restores mission completion at unchanged safety—an instructive demonstration that an uncertainty-aware margin is only as sensible as the lifetime of the estimate feeding it.

Table 4. Dynamic-obstacle results ($N_{MC} = 50$, crossing obstacle, intermittent visibility, track management active). Waypoints visited counts true passes within 0.15 m; miss distance is the mean true closest approach over the three waypoints.

Strategy	Collision (95% CI)	Worst Clear. [m]	Path [m]	Wps Visited	Miss [m]
oracle (clairvoyant)	0% [0.0, 7.1]	$+0.002$	6.95	3.00/3	0.05
static (ignore motion)	88% [76.2, 94.4]	-0.268	7.01	2.42/3	0.09
cv_fixed	44% [31.2, 57.7]	-0.179	7.29	2.48/3	0.08
cv_cov (cov-aware)	**0% [0.0, 7.1]**	$+$**0.210**	15.44	2.00/3	0.11

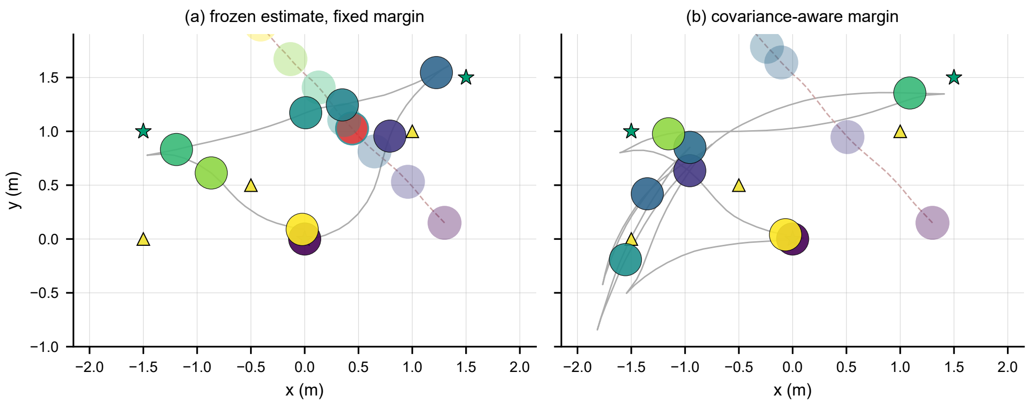

Figure 4. Dynamic obstacle under intermittent visibility, same trial (identical obstacle motion). Robot (solid) and obstacle (translucent) are drawn as disks ($r = 0.15$ m) at evenly spaced instants; simultaneous snapshots share a colour, and the obstacle disk turns red on overlap. (**a**) Frozen estimate with fixed margin: the disks overlap mid-crossing (collision). (**b**) Covariance-aware margin: the robot detours south of the coasting prediction and the simultaneous pairs never meet.

Figure 5 sweeps γ in the dynamic scenario. The collision rate collapses from 44% at $\gamma = 0$ to 2% at $\gamma = 0.5$ and is 0% for every $\gamma \geq 1$. Unlike the static case, the caution here is expensive—the path grows from 7.3 m to 15.7 m across the sweep, with a marked jump between $\gamma = 0.5$ (9.0 m) and $\gamma = 1$ (14.4 m)—because intermittent contact forces wide detours around coasting predictions. This suggests a simple deployment rule: where collisions are unacceptable and path budget is ample, adopt the *safety-first* default $\gamma = 2$, which sits well inside the collision-free plateau in both scenarios; where efficiency matters and a small residual risk is tolerable, the *efficient* setting $\gamma \approx 0.5$ removes 95% of the collisions (leaving $\approx 2\%$) at barely a quarter of the detour cost. The covariance scaling means either choice self-tunes to the live uncertainty; only the risk appetite is a design decision.

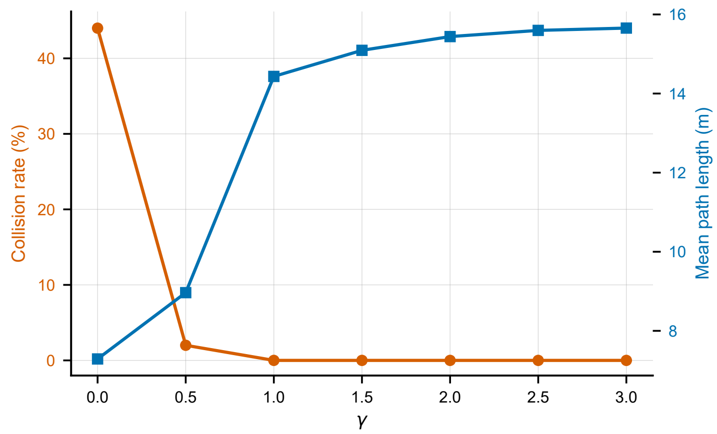

Figure 5. Dynamic obstacle: safety–efficiency trade-off under intermittent visibility. Collision rate (left axis) falls to the oracle floor as γ grows; mean path length (right axis) roughly doubles.

6.5. Ablation: Margin Size vs. Adaptivity

The covariance-aware margin fuses two effects: it is *larger* on average and it is *adaptive* (shaped by $\sqrt{\lambda_{\max}}$). To disentangle which of the two actually drives the safety gain, we run a matched-mean ablation. We first measure the mean inflation actually applied by the covariance-aware controller over all trials, then build a *fixed* margin equal to that mean, so both controllers apply the same average keep-out but the fixed one spreads it uniformly. Both are evaluated on identical seeds.

Table 5 reports the outcome for both scenarios. Against the sensed static obstacle, at matched mean margin (a constant inflation of 0.146 m, the mean the covariance-aware controller applied at $\gamma = 2$), the fixed control is *equally safe* (0 of 50 collisions): the collision-free level is therefore governed primarily by the average margin size, not by adaptivity itself. The covariance-aware shaping nonetheless retains two advantages. First, it sets the margin *automatically* from the online covariances, whereas the fixed control required knowing the correct constant a priori—obtainable here only by first running the covariance-aware controller and measuring it. Second, at equal safety it follows a slightly shorter path (6.99 vs. 7.00 m) by concentrating clearance where and when uncertainty is high.

The same test in the dynamic scenario requires care in interpretation, and strengthens the same conclusion. The covariance-aware controller's mean applied inflation over the mission is large (4.44 m)—the average is dominated by coasting episodes in which a freshly created track's velocity covariance is still unconverged, and, the obstacle constraint being softened by a slack, demands of this size act as a strong repulsion rather than a literal keep-out. A constant margin of this matched size is, as in the static case, equally safe (0/50) and slightly *less* efficient (path 16.2 vs. 15.4 m; waypoints 2.08 vs. 2.00 of 3; worst clearance $+0.72$ vs. $+0.21$ m—conservatism spent where it is not needed). The static conclusion therefore extends: safety is governed by margin size, while the covariance-aware shaping *finds* the operating point online—here a size no designer sweeping constants would plausibly hand-pick—and concentrates its conservatism where the uncertainty is. The Wilson 95% intervals also show that a 0/50 outcome still admits an upper bound near 7%, and that the small-margin baselines carry wide intervals at $N_{\mathrm{MC}} = 50$.

Table 5. Margin size vs. adaptivity ablation ($N_{\mathrm{MC}} = 50$). The fixed-matched control uses a constant margin equal to the covariance-aware controller's mean applied inflation in that scenario. Collision rate with Wilson 95% confidence interval.

Scenario	Controller	Collision Rate (95% CI)	Path [m]
Static	fixed δ_0	14.0% [7.0, 26.2]	6.93
	fixed-matched (0.146 m)	0.0% [0.0, 7.1]	7.00
	covariance-aware (adaptive)	**0.0% [0.0, 7.1]**	**6.99**
Dynamic	cv_fixed (δ_0)	44.0% [31.2, 57.7]	7.29
	fixed-matched (4.44 m)	0.0% [0.0, 7.1]	16.23
	cv_cov (adaptive)	**0.0% [0.0, 7.1]**	**15.44**

6.6. Robustness to Monte-Carlo Sample Size

To confirm that the collision results are not artifacts of the trial count, we repeated the principal studies at $N_{\mathrm{MC}} = 100$ (extending the same seed stream, so the first 50 realizations coincide with the primary study). Table 6 reports the collision rates with Wilson 95% confidence intervals. All conclusions are unchanged: the covariance-aware margin remains collision-free against both the sensed static obstacle and the dynamic obstacle (0/100, upper bound 3.7%, matching the oracle); the matched-fixed ablation still holds (0/100 at the same 0.146 m mean margin); and the free-space tracking gain grows slightly to 54% versus odometry (0.062 vs. 0.135 m). Point estimates shift only within the $M = 50$ intervals (for example, dynamic cv_fixed 44% $\to$ 42% and static slam-fixed 14% $\to$ 15%).

Table 6. Robustness check at $N_{\mathrm{MC}} = 100$: collision rate with Wilson 95% confidence interval. All findings of the $M = 50$ study are reproduced.

Scenario	Strategy	Collision Rate (95% CI)
Static	odom + fixed δ_0	36.0% [27.3, 45.8]
	slam + fixed δ_0	15.0% [9.3, 23.3]
	slam + covariance-aware	**0.0% [0.0, 3.7]**
	fixed-matched (size only)	0.0% [0.0, 3.7]
Dynamic	oracle (clairvoyant)	0.0% [0.0, 3.7]
	static (ignore motion)	89.0% [81.4, 93.7]
	cv_fixed	42.0% [32.8, 51.8]
	cv_cov (covariance-aware)	**0.0% [0.0, 3.7]**

6.7. Computational Performance

Per-step computation was measured against the controller sampling period on the unknown-map controllers. In free space the NMPC solves in 6.1 ms on average (95th percentile 8.3 ms) against a 50 ms period, an $\approx$12% duty cycle. The parameterised-obstacle controller, a larger multiple-shooting problem with nonconvex constraints and slack variables, averages 11.5 ms (95th percentile 14.9 ms) against a 200 ms period, an $\approx$6% duty cycle. The EKF-SLAM and obstacle-estimator updates cost well under a millisecond, and the covariance term in (7) is a single small eigenvalue problem; all are negligible. The framework therefore runs with roughly an order-of-magnitude real-time margin, and the covariance-aware safety mechanism adds essentially no computational cost.

7. Discussion

Taken together, the experiments convey a nuanced message. Under certainty equivalence, feeding SLAM into the loop sharpens accuracy and softens the severity of collisions, yet it does not, by itself, make the system safe—a sufficiently large obstacle margin does.

The ablation (Table 5) makes this plain: the collision-free level is dictated by the average margin size, since a fixed margin of the same mean proves equally safe. The distinctive merit of sizing the margin from the live covariances is therefore not a categorical gain in safety but two eminently practical ones. First, it self-tunes the keep-out online, sparing the designer from hand-picking a safe constant a priori—a constant that, in our experiments, was obtainable only by first running the covariance-aware controller and measuring what it did. Second, it concentrates clearance precisely where and when uncertainty runs high, which in the unknown-map setting makes the safety remarkably cheap: against the sensed static obstacle, zero collisions cost less than 1% of path. Dead reckoning can accomplish neither, for it lacks any trustworthy estimate of its own uncertainty.

The unknown-map setting also surfaced three lessons a fully known world conceals. First, an uncertainty-aware margin inherits the *honesty* of the filter that feeds it. EKF-SLAM is known to become overconfident—its covariance shrinking faster than the true error—as a consequence of linearizing about an uncertain estimate under weak observability [9,10]; we observed precisely this in stretches where a single landmark was in range—the heading is then weakly observable, and an unlucky initialization can leave the estimate biased well beyond what the covariance admits. When the filter is overconfident, the margin under-inflates in proportion. Second, as the dynamic study showed, the margin is only as sensible as the lifetime of the estimate feeding it: without track management, a coasting track's unbounded covariance converts caution into paralysis. Third, safety and perception are *coupled through the trajectory*: the evasive detours that buy safety also carry the robot away from its landmarks, degrading the very localization that mission precision depends on, so waypoints are attained in the belief frame while the true pass drifts wide (Table 4). The margin itself remains consistent under this self-induced degradation—it scales with the covariance that the detour grows—but the observation argues for perception-aware detouring, i.e., resolving avoidance ambiguity toward landmark coverage, as a natural refinement.

Several modeling choices bound the scope. Landmarks are assumed distinguishable (known data association), which suits uniquely identifiable features or beacons but sidesteps the association failures of cluttered environments. The margin (7) uses a Gaussian, single-obstacle tightening and assumes a reasonably calibrated filter; it is chance-constraint-inspired rather than a calibrated probability bound, and a formal bound would require accounting for linearization and correlation between obstacles. The study is in simulation with three landmarks and a single obstacle per scenario, with γ selected by sweeps on those same geometries—the collision-free operating points are demonstrated, not guaranteed, for unseen layouts—and the controller uses certainty equivalence for the tracking term (only the obstacle constraint is made uncertainty-aware). The sparse map is also a deliberately demanding case: landmark density governs how quickly the pose covariance grows between re-detections, so a denser map would blunt the very nonstationarity the margin responds to, while a sparser one would sharpen both the drift and the perception–control coupling reported below; a systematic density sweep is left to future work.

Natural extensions include multiple simultaneous obstacles, tube-MPC bounds for the tracking objective, consistency-preserving estimators as the margin's information source, hardware validation on a differential-drive or car-like platform, and a formal chance-constraint derivation that propagates the predicted covariance along the horizon.

8. Conclusions

This study has presented a closed-loop framework that couples EKF-SLAM with a constrained nonlinear MPC for a nonholonomic mobile robot navigating an environment that is unknown before it is sensed—landmarks initialized from first detection, obstacles discovered and estimated online—and, reaching beyond certainty equivalence, feeds the live covariances back into the controller to size the obstacle keep-out constraints. On a multi-leg patrol in Monte-Carlo simulation, SLAM-in-the-loop reduced terminal error by 48% relative to odometry while building the map from scratch to 4 cm accuracy, and the proposed covariance-aware safety margin eliminated collisions with a sensed static obstacle (0 of 50 trials) for less than 1% additional path—the margin spends clearance only in the landmark-poor passage where the uncertainty actually resides. An ablation clarified the mechanism: a fixed margin of the same average size is equally safe, so the collision-free level is governed by margin size, whereas the distinctive value of the covariance-aware formulation lies in self-tuning that margin online. A sweep of the scaling factor characterized the safety–efficiency trade-off and justified the operating point, and timing measurements confirmed real-time feasibility with an order-of-magnitude margin. The same covariance-aware mechanism extends to a moving, intermittently visible obstacle tracked by a coasting constant-velocity filter, where it matched the clairvoyant oracle (0% vs. 44–88% for fixed-margin strategies) at the price of a more cautious path and of some true-frame waypoint precision, since the evasive detours themselves degrade localization—and provided stale tracks are dropped, without which unbounded prediction uncertainty trades mission completion for safety. The results indicate that the uncertainty already computed by the filters is a valuable, underused signal for safe predictive control in unknown environments.

Author Contributions: Conceptualization, D.L. and L.W.; methodology, D.L.; software, D.L.; validation, D.L., L.W. and R.P.G.; formal analysis, D.L.; writing—original draft preparation, D.L.; writing—review and editing, L.W. and R.P.G.; supervision, L.W. and R.P.G. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Data Availability Statement: The MATLAB/CasADi code and simulation data supporting the reported results are available from the corresponding author upon reasonable request.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Acknowledgments: During the preparation of this manuscript, the authors used Anthropic Claude (Claude Code) for the purposes of software-code development, manuscript drafting and language editing, and figure-script generation. The authors have reviewed and edited the output and take full responsibility for the content of this publication.

Conflicts of Interest: The authors declare no conflicts of interest.

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